Shubham Kanodia

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Appointments

Carnegie Institution for Science — **Washington, DC, USA**
Carnegie Postdoctoral Fellow, Earth and Planets Laboratory — July 2022 – Present
From Pixels to Population: Understanding Gas Giants around M dwarfs

Pennsylvania State University — **Pennsylvania, USA**
Research Technologist — Feb 2017 – July 2017
HPF and NEID spectrograph design and instrument assembly

Education

Pennsylvania State University — **Pennsylvania, USA**
Doctor of Philosophy (Ph.D.) Astrophysics — May 2019 – May 2022
Developing new tools and techniques to probe the M dwarf planet population

Pennsylvania State University — **Pennsylvania, USA**
Master of Science (M.Sc.) Astrophysics — Sept 2017 – May 2019
Combining the Next Generation of Exoplanet Instrumentation & Astrostatistics

Brown University — **Rhode Island, USA**
Master of Science (Sc.M.) Physics — Sept 2015 – Dec 2016
Optical Design of the Exoplanet Climate Infrared Telescope Spectrometer

St. Xavier's College — **Mumbai, India**
Bachelor of Science (B.Sc.) Physics — June 2012 – Apr 2015

Telescope Time Allocation

- JWST PI: $>$ 150 hours
- HET 10 m HPF: $>$ 120 nights
- Gemini-N 8 m MAROON-X: 3.5 nights
- Magellan 6.5 m: $>$ 20 nights
- ARC 3.5 m: $>$ 50 half nights
- WIYN 3.5 m: $>$ 30 nights

Awards & Grants

I am the PI and/or Science PI of $\gtrsim$ \$1.8 million in awards/grants.

- NASA U.S. Contributions to Ariel Preparatory Science (Science PI), 2025-2027
- JWST GO programs (as PI), 2023-2027
- NN-Explore JPL PI Data Awards, 2021, 2023-2025
- Carnegie Postdoctoral Fellow, Carnegie Earth & Planets Lab, 2022 onwards
- NASA Group Achievement Award, NEID, 2020
- Downsbrough Graduate Fellowship in Astrophysics, Penn State, 2021
- Zaccheus Daniel Fellowship, Penn State, 2018, 2020, 2021
- Homer F. Braddock / Nellie H. and Oscar L. Roberts Fellowship, Penn State, 2017
- J.N. Tata Endowment Fund for Higher Education, Mumbai, 2015

Mentoring & Advising

- Helen Baran (2019 – 2020): Undergraduate at Pennsylvania State University
- Marissa Maney (2019 – 2021): Undergraduate at Pennsylvania State University
- Brody McElwain (2020 – 2022): Master's student at Pennsylvania State University
- Narisara (Mick) Mayer (2023): Undergraduate at Haverford College
- Caleb Dando-Haenisch (2023): Undergraduate at American University
- Radia Islam (2023): Undergraduate at University of Texas, Austin
- Amber Wong (2023 – 2025): Undergraduate at University of California, Irvine
- Shane O'Brien (2023 – 2026): Undergraduate at University of California, Irvine

- Andrew Hotnisky (2023 – 2025): Undergraduate at Pennsylvania State University
- Rowen Glusman (2023 – 2026): Master's student at University of Amsterdam
- Fogofoluwa Adeniyan (2024): Undergraduate at Brightpoint Community College
- Nachiket Yadav (2024 – 2026): Undergraduate at University of Texas, Austin
- Maya Vigil (2024): Undergraduate at University of Texas, Austin
- Gogod James (2025 –): Undergraduate at University of California, Irvine

Co-advised:

- Megan Delamer (2022 – 2026): PhD student at Pennsylvania State University
- Pinchen Fan (2022 – 2023): PhD student at Pennsylvania State University
- Lia Marta Bernabò (2023 – 2024): PhD student at Institute of Planetary Research - DLR (Germany)
- Varghese Reji (2023 – 2025): PhD student at Tata Institute of Fundamental Research (India)
- Te Han (2023): PhD student at University of California, Irvine
- Alexandra Boone (2024 –): PhD student at University of Wyoming
- Pranav Premnath (2025 –): PhD student at University of California, Irvine

Academic Service

- **Carnegie AI Working Group**, Member
- **Carnegie Inclusion Diversity Equity Accessibility (IDEA) Working Group**, Member
- **NASA Postdoctoral Program (NPP)**, Reviewer
- **STScI JWST Telescope Allocation Committee (TAC)**, Discussion Panelist (2×)
- **Canada Foundation for Innovation (CFI)**, Expert Reviewer
- **Canadian Time Allocation Committee (CanTAC)**, Expert Reviewer: Gemini, CFHT
- **NSF's NOIRLab Telescope Time Allocation Committee**, Chair, Panelist
- **NASA Exoplanets Research Program (XRP)**, Panelist, Executive Secretary
- **Nature, ApJ, AJ, MNRAS, A&A, RAS Techniques and Instruments, International Journal of Astrobiology**, Reviewer
- **Carnegie EPL Astro Seminars**, Organizer, 2023-2025
- **Emerging Researchers in Exoplanet Sciences IV, VII**, Organizing Committee, 2018, 2022

Teaching

Teaching probabilistic programming — **State College, USA**
Pennsylvania State University — 2021 and 2022
Developed and taught a course on astronomical computing (Python, Git, high-performance computing), probabilistic programming and statistical inference using the Hamiltonian Monte Carlo codes: `PyMC3` and package `exoplanet`.

Teaching Assistant for Astronomy lab — **Providence, USA**
Brown University — Jan 2016 – Apr 2016
Lab assistant for basic astronomy lab course.

Software Development

- `pyGaiafits`: Repository to fit Gaia epoch astrometry with a single-star or binary orbit model, optionally jointly with radial velocities (Kanodia et al., submitted).
- `exoComet`: Repository with a 1D Markov-chain model for comet delivery in compact planetary systems — collision, ejection and escape probabilities without N-body integration. (Chambers and Kanodia, submitted).
- `pyastrotools`: Repository with a set of astronomy helper functions
- `barycorrpy`: Python package for precise barycentric and timing corrections. (Kanodia and Wright, 2018; Wright and Kanodia, 2020).
- `MRExo`: Nonparametric tool used to fit 2-D, 3-D and 4-D mass- radius+ relationships using beta density functions. (Kanodia et al. 2019, 2023).

Outreach

In addition to regular Astronomy open nights and public events at each of my host institutions, I have contributed to the following:

- **Public Talks**:
  - Galesville Astrophysical Society (GAS): Washington DC, USA, 2026
  - Carnegie Neighbourhood Lecture: Washington DC, USA, 2025
  - National Capital Astronomers: Washington DC, USA, *A Gas-giant planet orbiting a Dwarf Star: An extreme Instance of Planet Formation*, 2024
  - Astronomy on Tap: State College, USA, *Digging through the Cosmic Haystack*, 2019
  - Nerd Nite: Webster's Cafe, State College, USA, *Searching for other worlds, other life*, 2019
  - Nehru Planetarium, Mumbai, India, *Finding Earth 2.0*, 2018
- **Education**:
  - Revamping and upgrading the astronomy exhibits at the US National Park Service's Rock Creek Park Nature Center and Planetarium, Washington DC to reflect latest discoveries and improve accessibility (2025)
  - Volunteered for Carnegie Academy for Science Education (CASE): First Light science program for middle-school kids (2022 – 23)
  - Volunteered for Astrofest: Penn State Department of Astronomy Annual outreach event (2017 – 2019)
  - Volunteered with Brown Cubesat Educational Outreach Saturday STEM program at West Broadway Middle School to communicate Science and Physics to students. (2015 – 2016)
  - Volunteered at Umang Foundation, Mumbai: Teaching underprivileged children basic Mathematics and English. (2012 – 2014)

Professional Talks

Invited Seminars & Colloquia

- **University of Wyoming Colloquium**, Wyoming, April 2026
- **NASA Goddard, Extrasolar Planets Seminar**, Maryland, January 2025
- **Pennsylvania State University, Special Seminar/Colloquium**, Pennsylvania, January 2025
- **ETH Zurich, Planetary Seminar**, Switzerland, June 2024
- **University of Zurich, Planetary Seminar**, Switzerland, June 2024
- **Carnegie Observatories, Special Seminar**, California, February 2024
- **University of Maryland, Planetary Seminar**, Maryland, November 2023
- **American University, Physics Colloquium**, District of Columbia, November 2023
- **Steward Observatory/NSF's NOIRLab Joint Colloquium**, Arizona, October 2023
- **Tata Institute of Fundamental Research, DAA Seminar**, India, August 2023
- **Carnegie EPL, General Seminar**, District of Columbia, February 2023
- **Tata Institute of Fundamental Research, DAA Seminar**, India, March 2022
- **Carnegie EPL, Astronomy Seminar**, District of Columbia, October 2021
- **Pennsylvania State University, Center for Exoplanets and Habitable Worlds Seminar**, Pennsylvania, September 2021
- **NASA Goddard, Extrasolar Planets Seminar**, Maryland, September 2021
- **Order of the Octopus Seminar**, Pennsylvania, July 2021
- **Pennsylvania State University, PSETI Seminar**, Pennsylvania, October 2020

Contributed Conference Presentations

- **Gas Giant Planet Formation Conference**, Michigan, May 2026
- **AASTCS 11: Exoplanet Atmospheres 2026**, Colorado, March 2026
- **Magellan Science Meeting**, District of Columbia, May 2025
- **American Astronomical Society, 245 Winter Meeting**, Maryland, January 2025

- **Open Problems in the Astrophysics of Gas Giants Conference**, Chile, December 2023
- **Strange New Worlds Conference**, India, August 2023
- **Origins of Solar Systems, Gordon Research Seminar**, New Hampshire, June 2023
- **NISER School of Earth and Planetary Sciences (SEPS) Seminar**, India, March 2023
- **SPIE Astronomical Telescopes + Instrumentation Conference**, Canada, August 2022
- **NASA Technosignatures Workshop**, Texas, September 2018
- **Emerging Researchers in Exoplanet Science Symposium**, Pennsylvania, June 2018

Selected Media & Press Coverage

- **Popular Science**, "Astronomers Found a 'Forbidden Planet' With a Strange Atmosphere," April 2026
- **Nautilus**, "Cosmic Mysteries Swirl Around 'Forbidden Planet'," April 2026
- **NDTV**, "Forbidden Planet the Size of Jupiter That Shouldn't Exist in Theory Baffles Scientists," April 2026
- **The Planetary Society: Planetary Radio**, "Exoplanet Enigma: Unpacking the Discovery of a 'Forbidden' Planet," May 2023
- **CNN**, "Astronomers Spot 'Forbidden' Planet That Shouldn't Exist Around a Tiny Star," February 2023
- **Space.com**, "Massive 'Forbidden Planet' Orbits a Strangely Tiny Star Only 4 Times Its Size," February 2023
- **The Independent**, "'Forbidden' World Could Change Our Understanding of How Planets Form," February 2023
- **ScienceDaily**, "'Forbidden' Planet Orbiting Small Star Challenges Gas Giant Formation Theories," February 2023
- **Phys.org**, "'Forbidden' Planet Orbiting Small Star Challenges Gas Giant Formation Theories," February 2023
- **EurekAlert!**, "'Forbidden' Planet Orbiting Small Star Challenges Gas Giant Formation Theories," February 2023
- **Space.com**, "'Marshmallow' World Defies Expectations for Planets Orbiting Red Dwarf Stars," October 2022
- **NSF NOIRLab**, "Astronomers Discover 'Marshmallow' World Orbiting a Cool Red Dwarf Star," October 2022
- **SETI Institute**, "Marshmallow Gas Giant Found at Red Dwarf Star," October 2022
- **Gizmodo**, "Planet With the 'Density of a Marshmallow' Spotted Around Red Dwarf Star," October 2022
- **Newsweek**, "Scorching Hot Planet the Size of Jupiter Is as Dense as a Marshmallow," March 2022
- **AAS Nova**, "Searching for Alien Needles in the Cosmic Haystack," November 2018
- **VICE**, "If the Universe Is the Ocean, We've Only Searched for Aliens in a Hot Tub," October 2018
- **Business Insider**, "Aliens Haven't Been Found Because We've Hardly Looked for Them: Study," October 2018
- **Smithsonian Magazine**, "In Search for Aliens, We've Only Examined a 'Cosmic Hot Tub'," October 2018

Publications

First Author (Refereed): 20 (14); Significant Contributions: 38 [ADS] 0000-0001-8401-4300
Total Citations: 2509 on 10 October, 2026. Mentee Publications are indicated with *.
Submitted and in prep. papers can be sent upon request.

First Author

20. **S. Kanodia**, C. Cañas, J. Lustig-Yaeger, et al., *GEMS JWST: Disentangling Stellar Contamination to reveal a Stellar Mass offset and Planet Mass trend with Atmospheric Metallicity*, In Prep.
19. **S. Kanodia**, S. Mahadevan, G. Stefansson, et al., *A Joint Study of Epoch Astrometry and Radial Velocities of Gaia-4b*, Submitted to AAS Journals.
18. **S. Kanodia**, C. Cañas, J. Lustig-Yaeger, et al., *GEMS JWST: A Subsolar Metallicity Atmosphere for Giant Planet TOI-5293 A b Orbiting a Rapidly Changing M Dwarf*, ApJ, 1006, 2, 2026 [ADS]
17. **S. Kanodia**, *Transiting Jupiters around M Dwarfs Have Similar Masses to FGK Warm Jupiters*, ApJ, 978, 1, 2025 [ADS]
16. **S. Kanodia**, C. Cañas, S. Mahadevan, et al., *Searching for GEMS: TOI-7149 b, an Inflated Giant Planet Causing a 12% Transit of a Fully Convective M-dwarf*, AJ, 170, 4, 2025 [ADS]
15. **S. Kanodia**, and A. Monson, *Precise Near-infrared Occultation of WASP-19b with FourStar on Magellan*, RNAAS, 9, 4, 2025 [ADS]
14. **S. Kanodia**, A. Gupta, C. Cañas, et al., *Searching for GEMS: Characterizing Six Giant Planets Around Cool Dwarfs*, AJ, 168, 6, 2024 [ADS]
13. **S. Kanodia**, C. Cañas, S. Mahadevan, et al., *Searching for Giant Exoplanets around M-dwarf Stars (GEMS) I: Survey Motivation*, AJ, 167, 4, 2024 [ADS]
12. **S. Kanodia**, A. Lin, E. Lubar, et al., *Stable Fiber-illumination for Extremely Precise Radial Velocities with NEID*, AJ, 166, 3, 2023 [ADS]
11. **S. Kanodia**, M. He, E. Ford, et al., *Beyond Two-dimensional Mass-Radius Relationships: A Nonparametric and Probabilistic Framework for Characterizing Planetary Samples in Higher Dimensions*, ApJ, 956, 2, 2023 [ADS]
10. **S. Kanodia**, S. Mahadevan, J. Libby-Roberts, et al., *TOI-5205b: A Short-period Jovian Planet Transiting a Mid-M Dwarf*, AJ, 165, 3, 2023 [ADS]
9. **S. Kanodia**, J. Libby-Roberts, C. Cañas, et al., *TOI-3757 b: A Low-density Gas Giant Orbiting a Solar-metallicity M Dwarf*, AJ, 164, 3, 2022 [ADS]
8. **S. Kanodia**, L. Ramsey, M. Maney, et al., *High-resolution Near-infrared Spectroscopy of a Flare around the Ultracool Dwarf vB 10*, ApJ, 925, 2, 2022 [ADS]
7. **S. Kanodia**, S. Halverson, J. Ninan, et al., *A Harsh Test of Far-field Scrambling with the Habitable-zone Planet Finder and the Hobby-Eberly Telescope*, ApJ, 912, 1, 2021 [ADS]
6. **S. Kanodia**, G. Stefansson, C. Cañas, et al., *TOI-532b: The Habitable-zone Planet Finder confirms a Large Super Neptune in the Neptune Desert orbiting a metal-rich M-dwarf host*, AJ, 162, 4, 2021 [ADS]
5. **S. Kanodia**, C. Cañas, G. Stefansson, et al., *TOI-1728b: The Habitable-zone Planet Finder Confirms a Warm Super-Neptune Orbiting an M-dwarf Host*, ApJ, 899, 1, 2020 [ADS]
4. **S. Kanodia**, J. Ninan, A. Monson, et al., *Ghosts of NEID's past*, SPIE, 11447, 2020 [ADS]
3. **S. Kanodia**, A. Wolfgang, G. Stefansson, et al., *Mass-Radius Relationship for M Dwarf Exoplanets: Comparing Nonparametric and Parametric Methods*, ApJ, 882, 1, 2019 [ADS]
2. **S. Kanodia**, and J. Wright, *Python Leap Second Management and Implementation of Precise Barycentric Correction (barycorrpy)*, RNAAS, 2, 1, 2018 [ADS]
1. **S. Kanodia**, S. Mahadevan, L. Ramsey, et al., *Overview of the spectrometer optical fiber feed for the habitable-zone planet finder*, SPIE, 10702, 2018 [ADS]

Significant Contributions

38. J. Chambers, and **S. Kanodia**, *Comet Delivery in Compact Planetary Systems Is Stellar-Mass Independent*, Submitted to AAS Journals.
37. R. Helled, S. Shibata, **S. Kanodia**, et al., *The metallicity of giant exoplanets around low-mass stars and how Ariel can help*, arXiv, Submitted to RASTI, 2026 [ADS]
36. G. Guzmán Caloca, C. Cañas, N. Wallack, et al., *GEMS JWST: Hold on to Your HATS(-6 b), a Subsolar-metallicity Giant Planet with Water, Methane, and Ammonia in its Atmosphere*, AJ, 172, 4, 2026 [ADS]
35. *A. Boone, H. Kobulnicky, C. Cañas, et al., *Searching for GEMS: Discovery of the Nearby Post-common-envelope Binary System TIC-460388167*, ApJ, 1003, 2, 2026 [ADS]
34. *R. Glusman, C. Cañas, **S. Kanodia**, et al., *Searching for GEMS: The Occurrence of Giant Planets Orbiting M Dwarfs within 100 pc*, AJ, 171, 3, 2026 [ADS]
33. A. Sandoval, C. Cañas, **S. Kanodia**, et al., *Searching for GEMS: TOI-5349 b is a Saturn-like Planet Orbiting a Metal-rich Early M Dwarf*, AJ, 171, 1, 2026 [ADS]
32. S. Logsdon, E. Hunting, **S. Kanodia**, et al., *Using on-sky, signal-to-noise triggered observations to monitor telescope mirror reflectivity performance*, Observatory Operations: Strategies, Processes, and Systems XI, 14151, 2026 [ADS]
31. R. Ashtari, J. Lustig-Yaeger, J. Libby-Roberts, et al., *GEMS JWST: HATS-75 b A Giant Planet with a Subsolar Metallicity Atmosphere Orbiting an M Dwarf*, AJ, 171, 5, 2026 [ADS]
30. *S. O'Brien, A. Wong, T. Han, et al., *Searching for GEMS: TOI-5916 b & TOI-6158 b are Two Saturn-density Planets Orbiting M2 Dwarfs*, AJ, 172, 2, 2026 [ADS]
29. C. Cañas, J. Lustig-Yaeger, S. Tsai, et al., *GEMS JWST: Transmission Spectroscopy of TOI-5205b Reveals Significant Stellar Contamination and a Metal-poor Atmosphere*, AJ, 171, 4, 2026 [ADS]
28. *P. Premnath, P. Robertson, **S. Kanodia**, et al., *Searching for GEMS: Three Warm Saturns and a Super-Jupiter Orbiting Four Early M Dwarfs*, AJ, 172, 2, 2026 [ADS]
27. T. Han, P. Robertson, T. Brandt, et al., *Hundreds of TESS Exoplanets Might Be Larger than We Thought*, The Astrophysical Journal Letters, 988, 1, 2025 [ADS]
26. R. Fernandes, **S. Kanodia**, M. Delamer, et al., *Searching for GEMS: Confirmation of TOI-5573 b, a Cool, Saturn-like Planet Orbiting an M Dwarf*, AJ, 170, 1, 2025 [ADS]
25. G. Stefánsson, S. Mahadevan, J. Winn, et al., *Gaia-4b and 5b: Radial Velocity Confirmation of Gaia Astrometric Orbital Solutions Reveal a Massive Planet and a Brown Dwarf Orbiting Low-mass Stars*, AJ, 169, 2, 2025 [ADS]
24. A. Larsen, T. Swaby, H. Kobulnicky, et al., *Searching for GEMS: Discovery and Characterization of Two Brown Dwarfs Around M Dwarfs*, AJ, 169, 5, 2025 [ADS]
23. *V. Reji, **S. Kanodia**, J. Ninan, et al., *Searching for GEMS: TOI-5688 A b, a Low-density Giant Orbiting a High-metallicity Early M-dwarf*, AJ, 169, 3, 2025 [ADS]
22. *A. Hotnisky, **S. Kanodia**, J. Libby-Roberts, et al., *Searching for GEMS: Two Super-Jupiters Around M Dwarfs that May have Formed via Gravitational Instability*, AJ, 170, 1, 2025 [ADS]
21. *M. Delamer, **S. Kanodia**, C. Cañas, et al., *TOI-4201: An Early M Dwarf Hosting a Massive Transiting Jupiter Stretching Theories of Core Accretion*, The Astrophysical Journal Letters, 962, 2, 2024 [ADS]
20. *L. Bernabò, **S. Kanodia**, C. Cañas, et al., *Searching for GEMS: TOI-6383Ab, a Giant Planet Transiting an M3-dwarf Star in a Binary System*, AJ, 168, 6, 2024 [ADS]
19. *T. Han, P. Robertson, **S. Kanodia**, et al., *TOI-5344 b: A Saturn-like Planet Orbiting a Super-solar Metallicity M0 Dwarf*, AJ, 167, 1, 2024 [ADS]
18. S. Jones, G. Stefánsson, K. Masuda, et al., *TOI-2015 b: A Warm Neptune with Transit Timing Variations Orbiting an Active Mid-type M Dwarf*, AJ, 168, 2, 2024 [ADS]

17. X. Wang, M. Rice, S. Wang, et al., *Single-star Warm-Jupiter Systems Tend to Be Aligned, Even around Hot Stellar Hosts: No Teff-Obliquity Dependency*, The Astrophysical Journal Letters, 973, 1, 2024 [ADS]
16. J. Libby-Roberts, M. Schutte, L. Hebb, et al., *An In-depth Look at TOI-3884b: A Super-Neptune Transiting an M4Dwarf with Persistent Starspot Crossings*, AJ, 165, 6, 2023 [ADS]
15. L. Powers, J. Libby-Roberts, A. Lin, et al., *TOI-3785 b: A Low-density Neptune Orbiting an M2-dwarf Star*, AJ, 166, 2, 2023 [ADS]
14. C. Cañas, **S. Kanodia**, J. Libby-Roberts, et al., *TOI-3984 A b and TOI-5293 A b: Two Temperate Gas Giants Transiting Mid-M Dwarfs in Wide Binary Systems*, AJ, 166, 1, 2023 [ADS]
13. S. Sheikh, **S. Kanodia**, E. Lubar, et al., *A Green Bank Telescope Search for Narrowband Technosignatures between 1.1 and 1.9 GHz During 12 Kepler Planetary Transits*, AJ, 165, 2, 2023 [ADS]
12. G. Stefánsson, S. Mahadevan, Y. Miguel, et al., *A Neptune-mass exoplanet in close orbit around a very low-mass star challenges formation models*, Science, 382, 6674, 2023 [ADS]
11. M. Lambert, C. Bender, **S. Kanodia**, et al., *TOI-5375 B: A Very Low Mass Star at the Hydrogen-burning Limit Orbiting an Early M-type Star*, AJ, 165, 5, 2023 [ADS]
10. A. Boss, and **S. Kanodia**, *Forming Gas Giants around a Range of Protostellar M-dwarfs by Gas Disk Gravitational Instability*, ApJ, 956, 1, 2023 [ADS]
9. C. Beard, P. Robertson, **S. Kanodia**, et al., *TOI-1696 and TOI-2136: Constraining the Masses of Two Mini-Neptunes with the Habitable-Zone Planet Finder*, AJ, 163, 6, 2022 [ADS]
8. C. Cañas, **S. Kanodia**, C. Bender, et al., *TOI-3714 b and TOI-3629 b: Two Gas Giants Transiting M Dwarfs Confirmed with the Habitable-zone Planet Finder and NEID*, AJ, 164, 2, 2022 [ADS]
7. A. Lin, A. Monson, S. Mahadevan, et al., *Observing the Sun as a Star: Design and Early Results from the NEID Solar Feed*, AJ, 163, 4, 2022 [ADS]
6. C. Beard, P. Robertson, **S. Kanodia**, et al., *GJ 3929: High-precision Photometric and Doppler Characterization of an Exo-Venus and Its Hot, Mini-Neptune-mass Companion*, ApJ, 936, 1, 2022 [ADS]
5. G. Stefánsson, R. Kopparapu, A. Lin, et al., *A Mini-Neptune and a Radius Valley Planet Orbiting the Nearby M2 Dwarf TOI-1266 in Its Venus Zone: Validation with the Habitable-zone Planet Finder*, AJ, 160, 6, 2020 [ADS]
4. J. Wright, and **S. Kanodia**, *Barycentric Corrections for Precise Radial Velocity Measurements of Sunlight*, PSJ, 1, 2, 2020 [ADS]
3. C. Cañas, G. Stefansson, **S. Kanodia**, et al., *A Warm Jupiter Transiting an M Dwarf: A TESS Single-transit Event Confirmed with the Habitable-zone Planet Finder*, AJ, 160, 3, 2020 [ADS]
2. C. Schwab, A. Monson, **S. Kanodia**, et al., *The NEID spectrometer: fibre injection system design*, SPIE, 11447, 2020 [ADS]
1. J. Wright, **S. Kanodia**, and E. Lubar, *How Much SETI Has Been Done? Finding Needles in the n-dimensional Cosmic Haystack*, AJ, 156, 6, 2018 [ADS]

Co-Author

57. P. Robertson, M. Endl, W. Cochran, et al., *A Revised Mass and Period for the Habitable Zone super-Earth GJ 3378 b: A Planet Straddling the Cosmic Shoreline*, ApJ, 1005, 1, 2026 [ADS]
56. G. Dransfield, M. Timmermans, D. Sebastian, et al., *MANGOS – II. Five new giant planets orbiting low-mass stars*, Monthly Notices of the Royal Astronomical Society, 547, 4, 2026 [ADS]
55. C. Murray, L. Garcia, B. Rackham, et al., *A Panchromatic JWST Spectrum of a Giant Starspot on the Fully Convective M Dwarf TOI-3884*, The Astrophysical Journal Letters, 1003, 2, 2026 [ADS]
54. D. Souto, I. Pascucci, K. Cunha, et al., *A Chemical Mismatch between Young Stars and Their Inner Disks*, The Astrophysical Journal Letters, 1001, 2, 2026 [ADS]

53. K. Taş, G. Stefansson, S. Fariz, et al., *An Earth-sized planet on a 5.4 h orbit around a nearby K dwarf*, Astronomy and Astrophysics, 707, 2026 [ADS]
52. S. Surana, C. Bender, C. Cañas, et al., *Stellar and Orbital Characterization of Three Low Mass M Dwarf Binary Stars with Dynamical Spectroscopy from the Habitable Zone Planet Finder*, AJ, 171, 3, 2026 [ADS]
51. J. Roth, J. Hartman, G. Bakos, et al., *The T16 Planet Hunt: 10,000 New Planet Candidates from TESS Cycle 1 and the Confirmation of a Hot Jupiter around TIC 183374187*, The Astrophysical Journal Supplement Series, 284, 1, 2026 [ADS]
50. M. Giovinazzi, E. Fitzmaurice, A. Gupta, et al., *The NEID Earth Twin Survey. IV. Confirming an 89 days, msini=10 Moplus Planet Orbiting a Nearby Sun-like Star*, AJ, 171, 5, 2026 [ADS]
49. J. Simon, J. Rodriguez, J. Yana Galarza, et al., *TOI-7169 b: A Hot Jupiter Transiting a Metal-poor Star*, AJ, 172, 2, 2026 [ADS]
48. Y. Wu, G. Xiao, R. Butler, et al., *Detection of Four Cold Jupiters through Combined Analyses of Radial Velocity and Astrometry Data*, AJ, 171, 3, 2026 [ADS]
47. E. Koo, G. Stefansson, R. Kavanagh, et al., *Spectroscopic characterization of LOFAR radio-emitting M dwarfs*, Astronomy and Astrophysics, 706, 2026 [ADS]
46. T. Han, P. Robertson, C. Cañas, et al., *NEIDSpecMatch: Stellar Parameter Estimation with NEID Spectra Using an Empirical Library*, RNAAS, 9, 3, 2025 [ADS]
45. L. Doyle, C. Cañas, J. Libby-Roberts, et al., *The First Spin-Orbit Obliquity of an M dwarf/brown dwarf system: an eccentric and aligned TOI-2119 b*, Monthly Notices of the Royal Astronomical Society, 536, 4, 2025 [ADS]
44. J. Dong, A. Chontos, G. Zhou, et al., *Origins of Super Jupiters: TOI-2145b has a Moderately Eccentric and Nearly Aligned Orbit*, AJ, 169, 1, 2025 [ADS]
43. M. Kunimoto, Z. Lin, S. Millholland, et al., *Two Earth-size Planets and an Earth-size Candidate Transiting the nearby Star HD 101581*, AJ, 169, 1, 2025 [ADS]
42. R. Rodríguez Martínez, J. Eastman, K. Collins, et al., *Discovery and Characterization of an Eccentric, Warm Saturn Transiting the Solar Analog TOI-4994*, AJ, 169, 2, 2025 [ADS]
41. M. Giovinazzi, C. Blake, P. Robertson, et al., *The NEID Earth Twin Survey. II. Dynamical Masses in Seven High-acceleration Star Systems*, AJ, 170, 1, 2025 [ADS]
40. J. Burt, R. Zellem, D. Ciardi, et al., *A New Approach to Compiling Exoatmospheric Target Lists and Quantifying the Ground-based Resources Needed to Vet Them*, AJ, 170, 4, 2025 [ADS]
39. A. Gupta, J. Luhn, J. Wright, et al., *The NEID Earth Twin Survey. I. Confirmation of a 31 Day Planet Orbiting HD 86728*, AJ, 169, 1, 2025 [ADS]
38. C. Beard, P. Robertson, J. Lubin, et al., *Discovery of a Nearby Habitable Zone Super-Earth Candidate Amenable to Direct Imaging*, AJ, 170, 5, 2025 [ADS]
37. A. Gupta, E. Fitzmaurice, S. Mahadevan, et al., *The NEID Earth Twin Survey. III. Survey Performance after Three Years on Sky*, AJ, 170, 5, 2025 [ADS]
36. E. Fitzmaurice, G. Stefánsson, R. Kavanagh, et al., *Astrometry and Precise Radial Velocities Yield a Complete Orbital Solution for the Nearby Eccentric Brown Dwarf LHS 1610 b*, AJ, 168, 3, 2024 [ADS]
35. M. Battley, K. Collins, S. Ulmer-Moll, et al., *NGTS-30b/TOI-4862b: An 1 Gyr old 98-day transiting warm Jupiter*, Astronomy and Astrophysics, 686, 2024 [ADS]
34. A. Alqasim, N. Grieves, N. Rosário, et al., *TOI-757 b: an eccentric transiting mini-Neptune on a 17.5-d orbit*, Monthly Notices of the Royal Astronomical Society, 533, 1, 2024 [ADS]
33. C. Beard, P. Robertson, M. Giovinazzi, et al., *Utilizing Photometry from Multiple Sources to Mitigate Stellar Variability in Precise Radial Velocities: A Case Study of Kepler-21*, AJ, 168, 4, 2024 [ADS]
32. A. Gupta, S. Millholland, H. Im, et al., *A hot-Jupiter progenitor on a super-eccentric retrograde orbit*, Nature, 632, 8023, 2024 [ADS]
31. E. Ford, C. Bender, C. Blake, et al., *Earths within Reach: Evaluation of Strategies for Mitigating Solar Variability using 3.5 years of NEID Sun-as-a-Star Observations*, arXiv, 2024 [ADS]

30. L. Zhao, X. Dumusque, E. Ford, et al., *The Extreme Stellar-signals Project. III. Combining Solar Data from HARPS, HARPS-N, EXPRES, and NEID*, AJ, 166, 4, 2023 [ADS]
29. J. Dong, S. Wang, M. Rice, et al., *TOI-1859b: A 64 Day Warm Jupiter on an Eccentric and Misaligned Orbit*, The Astrophysical Journal Letters, 951, 2, 2023 [ADS]
28. J. Lubin, X. Wang, M. Rice, et al., *TOI-1670 c, a 40 day Orbital Period Warm Jupiter in a Compact System, Is Well Aligned*, The Astrophysical Journal Letters, 959, 1, 2023 [ADS]
27. R. Frazier, G. Stefánsson, S. Mahadevan, et al., *NEID Reveals That the Young Warm Neptune TOI-2076 b Has a Low Obliquity*, The Astrophysical Journal Letters, 944, 2, 2023 [ADS]
26. A. Gupta, J. Jackson, G. Hébrard, et al., *A High-Eccentricity Warm Jupiter Orbiting TOI-4127*, AJ, 165, 6, 2023 [ADS]
25. A. Lin, J. Libby-Roberts, J. Alvarado-Montes, et al., *The Unusual M-dwarf Warm Jupiter TOI-1899 b: Refinement of Orbital and Planetary Parameters*, AJ, 166, 3, 2023 [ADS]
24. C. Cañas, S. Mahadevan, C. Bender, et al., *An Eccentric Brown Dwarf Eclipsing an M dwarf*, AJ, 163, 2, 2022 [ADS]
23. A. Gupta, C. Bender, J. Ninan, et al., *Real-time exposure control and instrument operation with the NEID spectrograph GUI*, SPIE, 12189, 2022 [ADS]
22. S. Logsdon, M. Wolf, D. Li, et al., *The NEID port adapter: on-sky performance*, SPIE, 12184, 2022 [ADS]
21. C. Cañas, S. Mahadevan, W. Cochran, et al., *A Hot Mars-sized Exoplanet Transiting an M Dwarf*, AJ, 163, 1, 2022 [ADS]
20. J. Dong, C. Huang, G. Zhou, et al., *NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star*, The Astrophysical Journal Letters, 926, 2, 2022 [ADS]
19. M. Reefe, R. Luque, E. Gaidos, et al., *A Close-in Puffy Neptune with Hidden Friends: The Enigma of TOI 620*, AJ, 163, 6, 2022 [ADS]
18. A. Gupta, J. Luhn, J. Wright, et al., *Detection of p-mode Oscillations in HD 35833 with NEID and TESS*, AJ, 164, 6, 2022 [ADS]
17. G. Stefànsson, S. Mahadevan, C. Petrovich, et al., *The Warm Neptune GJ 3470b Has a Polar Orbit*, The Astrophysical Journal Letters, 931, 2, 2022 [ADS]
16. R. Terrien, A. Keen, K. Oda, et al., *Rotational Modulation of Spectroscopic Zeeman Signatures in Low-mass Stars*, The Astrophysical Journal Letters, 927, 1, 2022 [ADS]
15. A. Ghosh, S. Sharma, J. Ninan, et al., *Gaia 20eae: A Newly Discovered Episodically Accreting Young Star*, ApJ, 926, 1, 2022 [ADS]
14. J. Lubin, P. Robertson, G. Stefansson, et al., *Stellar Activity Manifesting at a One-year Alias Explains Barnard b as a False Positive*, AJ, 162, 2, 2021 [ADS]
13. S. Vissapragada, G. Stefánsson, M. Greklek-McKeon, et al., *A Search for Planetary Metastable Helium Absorption in the V1298 Tau System*, AJ, 162, 5, 2021 [ADS]
12. A. Gupta, J. Wright, P. Robertson, et al., *Target Prioritization and Observing Strategies for the NEID Earth Twin Survey*, AJ, 161, 3, 2021 [ADS]
11. S. Mahadevan, G. Stefánsson, P. Robertson, et al., *The Habitable-zone Planet Finder Detects a Terrestrial-mass Planet Candidate Closely Orbiting Gliese 1151: The Likely Source of Coherent Low-frequency Radio Emission from an Inactive Star*, The Astrophysical Journal Letters, 919, 1, 2021 [ADS]
10. V. Krishnamurthy, T. Hirano, G. Stefánsson, et al., *Nondetection of Helium in the Upper Atmospheres of TRAPPIST-1b, e, and f*, AJ, 162, 3, 2021 [ADS]
9. A. Roy, S. Halverson, S. Mahadevan, et al., *Solar Contamination in Extreme-precision Radial-velocity Measurements: Deleterious Effects and Prospects for Mitigation*, AJ, 159, 4, 2020 [ADS]
8. P. Robertson, G. Stefansson, S. Mahadevan, et al., *Persistent Starspot Signals on M Dwarfs: Multiwavelength Doppler Observations with the Habitable-zone Planet Finder and Keck/HIRES*, ApJ, 897, 2, 2020 [ADS]

7. G. Stefansson, C. Cañas, J. Wisniewski, et al., *A Sub-Neptune-sized Planet Transiting the M2.5 Dwarf G 9-40: Validation with the Habitable-zone Planet Finder*, AJ, 159, 3, 2020 [ADS]
6. G. Stefansson, S. Mahadevan, M. Maney, et al., *The Habitable Zone Planet Finder Reveals a High Mass and Low Obliquity for the Young Neptune K2-25b*, AJ, 160, 4, 2020 [ADS]
5. J. Ninan, G. Stefansson, S. Mahadevan, et al., *Evidence for He I 10830 Å Absorption during the Transit of a Warm Neptune around the M-dwarf GJ 3470 with the Habitable-zone Planet Finder*, ApJ, 894, 2, 2020 [ADS]
4. P. Robertson, T. Anderson, G. Stefansson, et al., *Ultrastable environment control for the NEID spectrometer: design and performance demonstration*, JATIS, 5, 2019 [ADS]
3. A. Metcalf, T. Anderson, C. Bender, et al., *Stellar spectroscopy in the near-infrared with a laser frequency comb*, Optica, 6, 2, 2019 [ADS]
2. J. Ninan, C. Bender, S. Mahadevan, et al., *The Habitable-Zone Planet Finder: improved flux image generation algorithms for H2RG up-the-ramp data*, SPIE, 10709, 2018 [ADS]
1. G. Stefansson, S. Mahadevan, L. Hebb, et al., *Toward Space-like Photometric Precision from the Ground with Beam-shaping Diffusers*, ApJ, 848, 1, 2017 [ADS]